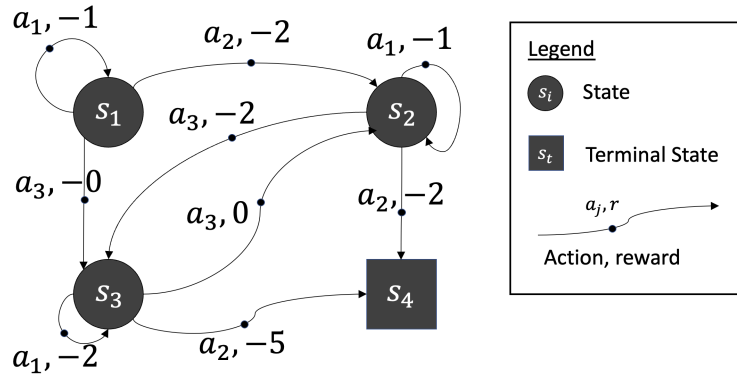


Exercise A



Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π .

- Provide the pseudo-code for the Iterative Policy Evaluation algorithm to estimate V_π in the generic MDP; in the algorithm consider
 - having reached K iterations as unique exit condition;
 - the 'in-place' version of the algorithm;
 - cycle over the states in a fixed ordered way ($s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_4$).
- Consider the deterministic MDP reported in the Figure above and consider a random policy π , ie:

$$\pi(a_1, s_j) = 0; \quad \pi(a_2, s_j) = \pi(a_3, s_j) = 0.5 \quad \forall s_j \in \mathcal{S}.$$

Estimate V_π by using iterative policy evaluation with:

- $\gamma = 1$;
- initial estimation $V_\pi = (0, 0, 0, 0)$;
- consider the 'in-place' version of iterative policy evaluation;
- stop the algorithm after reaching $K = 3$ as exit condition.

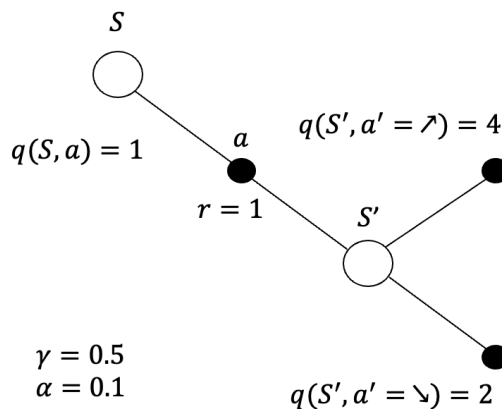
Report the procedure with calculations [hint: after 3 iterations, the estimation should be $V_\pi = (-\frac{81}{16}, -\frac{67}{16}, -\frac{147}{32}, 0)$].

- Report the pseudo-code of the Policy Iteration algorithm (using iterative policy evaluation): for the Policy Evaluation step, just refer to the algorithm reported at point 1.
- Consider the 'true' $V_\pi = (-\frac{11}{2}, -\frac{13}{3}, -\frac{14}{3}, 0)$: by applying one step of policy improvement derive a new deterministic policy π' .
- Consider a uniform random policy b , ie:

$$b(a_i, s_j) = 1/3; \quad \forall a_i \in \mathcal{A}, \forall s_j \in \mathcal{S}.$$

Consider the 'true' $V_b = (-\frac{15}{2}, -\frac{17}{3}, -\frac{19}{3}, 0)$: by applying one step of policy improvement derive a new deterministic policy b' . Is b' equal to π' derived at point 4?

Exercise B



- Provide the pseudo-code and backup diagram for the SARSA algorithm (on-policy TD control);
- Provide the pseudo-code and backup diagram for the Q-learning algorithm (off-policy TD control). Highlight (in the pseudocode) the differences between SARSA and Q-learning;

- REPORT YOUR ANSWERS TO EXERCISE A AND EXERCISE B IN THE PROTOCOL SHEET

Answer the following three questions:

- REPORT YOUR ANSWER TO QUESTION 1 IN THE FOLLOWING AND IN THE NEXT PAGES

[illegible]

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

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